Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/29/2017
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	11/30/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	• ISS increment 51 co-I (SCE [Zr/ZrNi]/TiZrNi), MUSC Control Center DLR Köln DE, 29-31 May, conducted MSL-EML flight space testing with Georg Lohöfer of DLR to investigate resistivity measurement protocols and with Dirk Holland-Moritz of DLR to evaluate convection influence on transformations in TiZrNi to compare to ground-based ESL results. • ISS expedition 52 co-I (Ge) MUSC Control Center DLR Köln DE, 24-25 July, conducted investigation on potential systematic contamination of gas atmosphere in space EML with Bernd Damaschke/Konrad Samwer of Georg-August Uni Gottingen. • Conducted ground-based electrostatic levitation (ESL) flight support testing at MSFC (Marshall Space Flight Center) January, March, and May 2017 on future space CoSi and FeCrNi alloys. • Purchased new high-speed digital camera for ground support testing. • Attended IWG#16 28-29 August at ESA-ESTEC. • Attended ECTP 2017 Thermophysical Property conference as invited speaker and session chairman at TU-Graz. • Attended ASGSR (American Society for Gravitational and Space Research) 2017 Annual conference as symposium organizer and presenter--October 2017.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Rodriguez JE, Matson DM. "Lateral heat flux and remelting during growth into the mushy-zone," Acta Materialia. 2017 May 1;129:408-14. https://doi.org/10.1016/j.actamat.2017.02.078 , May-2017
Articles in Peer-reviewed Journals	Matson DM, Xiao X, Rodriguez JE, Lee J, Hyers RH, Shuleshova O, Kaban I, Schneider S, Karrasch C, Burggraff S, Wunderlich R, Fecht H-J. "Use of thermophysical properties to select and control convection during rapid solidification of steel alloys using electromagnetic levitation on the Space Station." JOM (Journal of The Minerals, Metals & Materials Society). 2017 Aug;68(8):1311-8. https://doi.org/10.1007/s11837-017-2396-5 , Aug-2017
Articles in Peer-reviewed Journals	Matson DM, Hyers RW, Genau A, Asle Zaeem M. "Solidification behavior in reduced gravity." JOM (Journal of The Minerals, Metals & Materials Society). 2017 Aug;69(8):1258-60. https://doi.org/10.1007/s11837-017-2418-3 , Aug-2017

Articles in Peer-reviewed Journals	Jeon S, Matson DM. "Formation of cellular structure on metastable solidification of undercooled eutectic CoSi 62 at. %." Crystals. 2017 Oct;7(10):295. https://doi.org/10.3390/cryst7100295 , Oct-2017
Articles in Peer-reviewed Journals	Wang Y, Gao J, Kolbe M, Chuang A, Ren Y, Matson DM. "Metastable solidification of hypereutectic Co2Si-CoSi composition: microstructural studies and in-situ observations." Acta Materialia. 2018 Jan 1;142:172-80. Available online 18 September 2017. https://doi.org/10.1016/j.actamat.2017.09.037 , Jan-2018
Dissertations and Theses	Barlow S. "Investigation of Solidification Pathways in Co-Si Alloys." BS Honors Thesis, Mechanical Engineering Dept., Tufts University, March 2017. , May-2017
Dissertations and Theses	Xiao X. "Modeling of induced surface oscillation damping behavior in levitated molten metal droplets." PhD Dissertation, Mechanical Engineering Dept., Tufts University, August 2017. , Aug-2017